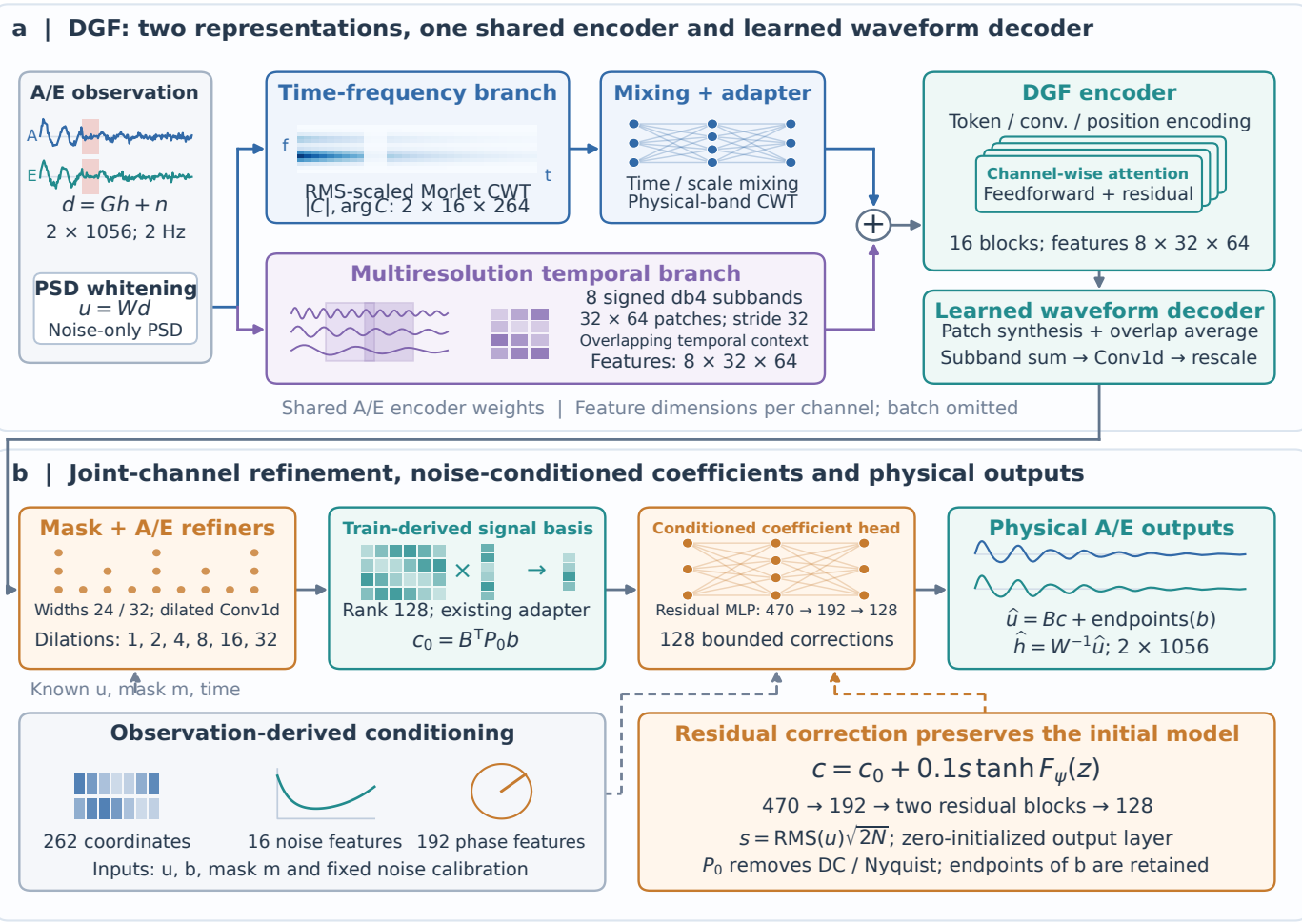

a | DGF: two representations, one shared encoder and learned waveform decoder
A/E observation
A
E
d = Gh + n
2 × 1056; 2 Hz
PSD whitening
u = Wd
Noise-only PSD
Time-frequency branch
f
t
RMS-scaled Morlet CWT
|C|, arg C: 2 × 16 × 264
Mixing + adapter
Time / scale mixing
Physical-band CWT
Multiresolution temporal branch
8 signed db4 subbands
32 × 64 patches; stride 32
Overlapping temporal context
Features: 8 × 32 × 64
DGF encoder
Token / conv. / position encoding
Channel-wise attention
Feedforward + residual
16 blocks; features 8 × 32 × 64
Learned waveform decoder
Patch synthesis + overlap average
Subband sum → Conv1d → rescale
Shared A/E encoder weights | Feature dimensions per channel; batch omitted
b | Joint-channel refinement, noise-conditioned coefficients and physical outputs
Mask + A/E refiners
Widths 24 / 32; dilated Conv1d
Dilations: 1, 2, 4, 8, 16, 32
Train-derived signal basis
Rank 128; existing adapter
c0 = B^T P0 b
Conditioned coefficient head
Residual MLP: 470 → 192 → 128
128 bounded corrections
Physical A/E outputs
û = Bc + endpoints(b)
ĥ = W^-1 û; 2 × 1056
Known u, mask m, time
Observation-derived conditioning
262 coordinates
16 noise features
192 phase features
Inputs: u, b, mask m and fixed noise calibration
Residual correction preserves the initial model
c = c0 + 0.1s tanh Fψ(z)
470 → 192 → two residual blocks → 128
s = RMS(u)√2N; zero-initialized output layer
P0 removes DC / Nyquist; endpoints of b are retained